Who Gets a Recycling Point?

Comparing collection-point types across Hong Kong's 18 districts

Group XX | DSC1001/GE1356 Introduction to Data Science | Option 2: Analyse Data Solutions
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1. Problem and questions

Hong Kong landfills **10,510 t** of waste a day. Households recycle only **22%** against a **≈55%** target, and residents say a nearby bin is their main reason to recycle (Ma et al., 2023). **How evenly does the network reach the 18 districts, and how do the three government solutions compare?**

RQ1. How uneven is access between districts?
RQ2. How do solutions A, B, C compare?
RQ3. Does access follow income, and why?
RQ4. Does access relate to waste or recycling?

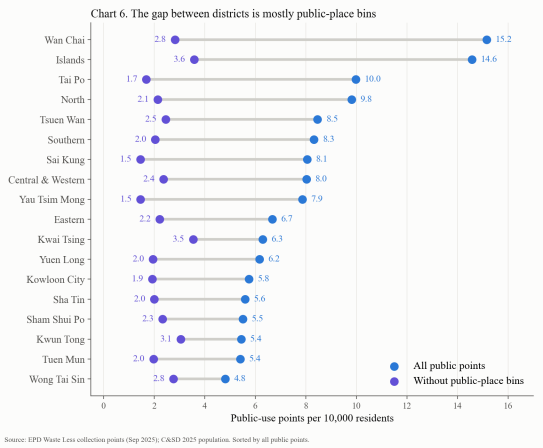
A public-place bins · **B** GREEN@COMMUNITY · **C** smart bins

2. Data and method

Data: EPD collection points (8,856 after cleaning; 5,301 public; Sep 2025); C&SD 2025 population, income and 2021 housing mix; EPD waste statistics 2022–24; 15 reports and papers in common units.

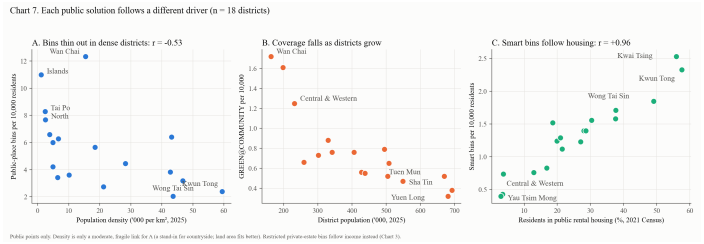
Method: points per 10,000 residents ($n = 18$ districts); Pearson r and partial correlations; six robustness checks incl. leave-one-out and a permutation test. Aggregated official data only; no personal data.

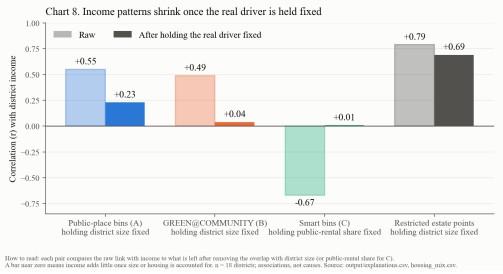
3. Who looks well served depends on what you count



Public points vary **3.2-fold** (Wong Tai Sin 4.8 to Wan Chai 15.2), driven by **public-place bins** in parks, offices and hiking roads. Without them, Kwai Tsing, Kwun Tong and Wong Tai Sin rise to the top.

4. Each solution follows a different driver





Public-place bins thin out in dense, urban districts (A). Income patterns shrink once **district size** (A, B) or **public rental housing** (C) is held fixed. Districts get a similar number of GREEN@COMMUNITY points, so the biggest get the least per resident.

5. Three solutions: no single winner

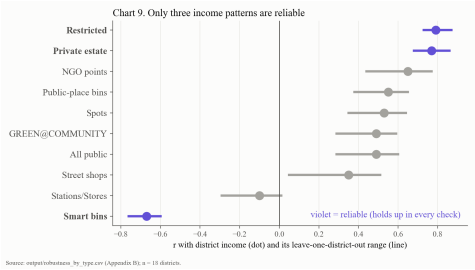
Chart 5. Three recycling solutions compared: no single winner

	A. Public-place bins	B. GREEN@COMMUNITY	C. Smart bins
Coverage public points per 10,000 residents	4.79	0.65	1.43
Materials accepted kinds per point	2.6	8.4	1.2
Evenness across districts lower = more even	0.52	0.50	0.43
Opening hours	*Not stated (open-air)*	Stores 10-13 h a day; Spots 3 h a week	80% open 7 am-midnight
Who it reaches	Parks, offices, countryside	Thinnest in the biggest districts	Public rental estates
Tonnes collected a year	*Not published*	47,500	up to 20,000 (food waste)
Cost per tonne	*Not published*	HK$10,100	*Not published*

Darker = better on that measure (rows 1-3). Grey = no published figure. Compare within a row, not across rows.
Source: EPD Waste Less collection points (Sep 2025; public points A = 3,588, B = 491, C = 1,073); C&SD 2025 population; LegCo LCQ2 (2026) and Panel paper (2024). Tonnes and cost are not like-for-like.

A widest coverage, few materials. **B** most materials and the only published cost (HK$10,100/t), thinnest coverage. **C** reaches public estates, mostly food waste.

6. Evaluation



Every result keeps its direction, but only **3 of 20** tests are reliable. RQ4 stays **unanswered**: EPD district waste data are "indicative only".

7. Limitations

- Counts ignore capacity and hours (a 3-hour weekly Spot = a Station)
- One-day snapshot; $n = 18$; Wan Chai is influential
- "Public" is the EPD's label; no walking-distance measure
- Associations, not causes (ecological fallacy)

8. Conclusions and impact

1. **Measure access by type**, not total count
2. **Scale GREEN@COMMUNITY to population**: ≈59 more Spots lift Yuen Long, Sha Tin and Tuen Mun to the median
3. **Target Stores at unmanaged older private buildings**
4. **Publish output and cost** for every solution

References and member responsibilities

References: EPD (2025) *Waste Less collection points, Waste statistics 2024*; C&SD (2022, 2026) district tables; Environment Bureau (2021) *Waste Blueprint 2035*; HKSAR Government (2026) LCQ2; Chan et al. (2024); Ma et al. (2023); Guo et al. (2018); Zhao et al. (2025); DiGiacomo et al. (2018); Li et al. (2020); Yau (2012). Full list in report. **Responsibilities:** [Member 1: . . .] · [Member 2: . . .] · [Member 3: . . .] · [Member 4: . . .] · [Member 5: . . .] · [Member 6: . . .]